

6.2 Bond Alphabet Soup

A Bond is a security, issued by a government or corporation, that promises certain payments at future dates.

Time of last payment $\rightarrow$ maturity date
(redemption date)

Time of purchase $\rightarrow$ Issue date

special cases

1. Fixed Maturity date : Noncallable
2. One single pmt at maturity : Zero Coupon Bonds

(1.) Printed on a bond is its FACE VALUE F
(par value)

↳ the number used to calculate the size of coupon payments.

(2.) Nominal rate of α convertible m
times per year

↳ m is the # of coupons each year

(3.) $r = \frac{\alpha}{m}$ effective rate for the
coupon period.

The amount of each coupon payment is

$$F_r = \frac{F \alpha}{m}$$

(4.) The number of coupon periods is n

↳ so for a N -year bond $n = Nm$

(5.) The redemption amount at the end of n coupon periods is C

↳ amount the bond-holder receives at maturity excluding the regular coupon amount.

If $F = C$ we say the bond is a par-value bond or redeemable at par.

If no redemption is specified assume par-value bond.

(6.) The modified coupon rate g is

$$g = \frac{F_r}{C}$$

coupon amount is $Cg = Fr$

(7.) The annual yield rate will be i
The yield rate per coupon period is j

$$i = (1 + j)^m - 1$$

If the yield rate is expressed as a nominal rate I then $j = \frac{I}{m}$ and $i = (1 + \frac{I}{m})^m - 1$

(8.) We use G to denote the coupon amount in terms of j . We call G the base amount.

$$G = \frac{Fr}{j}$$

↳ amount you would need to invest now to create a perpetuity paying an amount of interest at the end of each coupon period = coupon amount.

$$Fr = Gj = Cg$$

(9.) we let P denote the price paid for the bond to yield j per interest period

(10.) let K be the value of the redemption at the issue date

$$K = C v_j^n \quad \text{where } v_j = \frac{1}{1+j}$$

purchaser pays P at $t=0$ and gets Fr at times $t=1, 2, \dots, n$ as well as a redemption value of C at time $t=n$.

BASIC PRICE FORMULA

$$\begin{aligned} P &= (Fr) a_{\overline{n}|j} + C v_j^n \\ &= (Fr) a_{\overline{n}|j} + K \end{aligned}$$

time 0 equation of value.

TYPE	LETTER	MEANING
extent	N	number of years in the bond term
frequency	m	number of coupons per year
extent	n	number of coupons, $n = Nm$
Coupon rate	α	nominal coupon rate, annual coupons total $F\alpha$
Coupon rate	r	coupon rate per coupon period, amount of each coupon is Fr
Coupon rate	g	modified coupon rate, coupon amount is Cg
Yield rate	I	nominal yield rate convertible m times per year
Yield rate	j	effective yield rate for the coupon period
Yield rate	i	annual effective yield rate
Dollar amount	F	face (or par) value, coupon amount is Fr
Dollar amount	C	redemption amount
Dollar amount	G	base amount, coupon amount is Gj
Dollar amount	P	price at issue
Dollar amount	K	present value of the redemption amount

TABLE (6.2.4) Bond variables

Problem: An eight-year \$3,000 10% bond with semiannual coupons and redemption value \$2,800 is bought at a price to give the investor an annual effective yield rate of 12%. Find the base amount and the price of the bond.

Solution. Find G and P

Given: $N=8$, $m=2$, $n=16$, $\alpha=10\%$.

$$r = \frac{10\%}{2} = 5\%, \quad C = 2800, \quad i = 12\%.$$

$$j = (1.12)^{1/2} - 1, \quad F = 3000$$

$$\begin{aligned} P &= 3000(.05) a_{\overline{16}|j} + 2800(1+j)^{-16} \\ &= 2664.61 \end{aligned}$$

$$G = \frac{3000(-.05)}{j} = 2572.88$$

Problem: A twelve-year \$2,000 8% par-value bond with quarterly coupons is bought for \$2,200. Find the effective yield rate per coupon period j , the annual effective yield i , and the nominal yield rate I convertible quarterly.

Solution. Find i, j, I

Given: $F = 2000$, $\alpha = .08$, $m = 4$, $P = 2200$
 $N = 12$, $n = 48$, $r = \frac{.08}{4} = .02$

$$2200 = \frac{2000(.08)}{4} a_{\overline{48}|j} + C v_j^{48}$$

$$2200 = 40 \left(\frac{1 - v_j^{48}}{j} \right) + 2000 v_j^{48}$$

$$j = .0169395 \Rightarrow I = .067758 \quad i = .0694992$$

Problem: A nine-year \$5,000 7% bond with semiannual coupons is purchased for \$4,986. This price allows the purchaser an annual effective yield of 6%. Calculate the redemption amount.

Solution. $N = 9$, $F = 5000$, $\alpha = .07$, $m = 2$, $P = 4986$
 $i = .06$

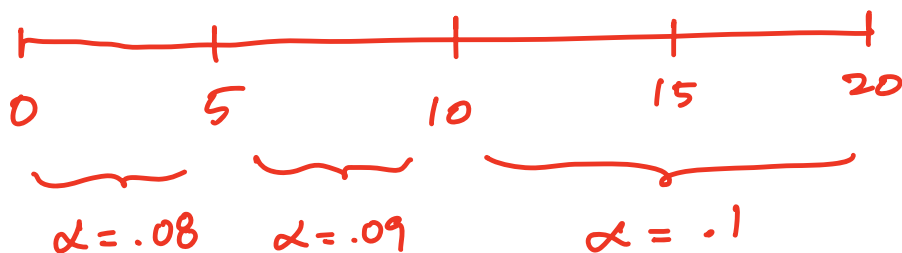
$$j = (1.06)^{1/2} - 1 \quad n = 9 \cdot 2 = 18$$

$$4986 = \frac{5000(.07)}{2} a_{\overline{18}|j} + C(1+j)^{-18}$$

$$C = 4342.33$$

Problem: A twenty-year \$1,000 bond with semiannual coupons is redeemable at par and has a nominal coupon rate of 8% for the first five years, 9% for the next five years, and 10% for the final ten years. Amy Astacio purchases the bond so as to receive a nominal yield of 9.2%. How much does she pay?

Solution. $N = 20$, $F = C = 1000$, $m = 2$, $I = .092$



$$j = \frac{.092}{2} = .046$$

$$n = 40$$

Coupons from $t=1$ to $t=10 = 40$

Coupons from $t=1$ to $t=10 = 45$

Coupons from $t=1$ to $t=10 = 50$

$$P = 40a_{\overline{40}|j} + 45(1+j)^{-10}a_{\overline{10}|j} + 50(1+j)^{-20}a_{\overline{20}|j} +$$

$$1000(1+j)^{-40}$$

$$= 968.72$$

Problem: Sydney and Sons Tool Corporation has structured its bond offering to appeal to the inflation conscious investor. This is done projecting an annual rate of inflation of 2.5%. An eight-year \$1,000 face value bond with annual coupons has a redemption amount equal to $\$1,218.40 \approx \$1,000(1.025)^8$, an initial coupon of \$80, and coupons that increase by 2.5% annually. How much should an investor who wishes to earn a noninflation-adjusted rate of 8% be willing to pay for the bond? If the actual rate of inflation is 2%, what is the investor's inflation-adjusted yield?

Solution. $N = 8, F = 1000, m = 1, C = 1218.40 = 1000(1.025)^8$
 coupons: $80, 80(1.025), 80(1.025)^2, \dots$ $i = 8\%$.

$$P = 80 \left(\frac{1 - \left(\frac{1.025}{1.08} \right)^8}{.08 - .025} \right) + 1218.40(1.08)^{-8}$$

$$= 1155.33$$

6.3 The Premium-Discount Formula

remember the basic price of a bond is :

$$\begin{aligned} P &= (Fr) a_{\overline{n}|j} + C v_j^n \\ &= (Fr) a_{\overline{n}|j} + K \end{aligned}$$

Since v_j^n appears in our annuity formula

$$a_{\overline{n}|j} = \frac{1 - v_j^n}{j} \Rightarrow v_j^n = 1 - j a_{\overline{n}|j}$$

Assuming $Fr = Cg$

So:

$$P = (Fr) a_{\overline{n}|j} + C v_j^n$$

$$P = (Cg) a_{\overline{n}|j} + C(1 - j a_{\overline{n}|j})$$

$$P = C(g - j) a_{\overline{n}|j} + C$$

this equation is known as the
"premium-discount" pricing formula
(very useful later on! Ammortization)

def: A bond is sold at a premium
if $P - C > 0$. We call $P - C$ the
premium or amount of premium.

$$\text{Premium} = P - C = C(g - j) a_{\overline{n}|j}$$

when $j < g$ the coupons Cg overpay
interest on the redemption amount C .

To make up for this, you buy the bond
at a premium.

def A bond is sold at a discount if $P - C < 0$. We let $C - P$ be the amount of discount

$$\text{discount} = C - P = C(j - g) a_{\overline{n}|i}$$

When $j > g$ the coupons Cg underpay interest on the redemption amount Cg . To compensate for this, you buy the bond at a discount.

Problem: A bond with a face value of \$6,000 and an annual coupon rate of 12% convertible semiannually will mature in ten years for its face value. If the bond is priced using a nominal yield rate of 6% convertible semiannually, what is the amount of premium or discount in this bond?

Solution. $F = C = 6,000$, $\alpha = 12\%$, $g = r = \frac{.12}{2} = .06$
 $N = 10$, $m = 2$, $n = 20$, $I = 6\%$, $j = 3\%$

$g > j$ so it's premium

$$\text{premium} = 6000(.06 - .03) a_{\overline{20}|.03} = 2677.95$$

Problem: A $7\frac{1}{2}$ -year 14% bond with a face value of \$2,500 has semiannual coupons and is sold to yield 7.2% convertible semiannually. The discount on the bond is \$283.12. Find the price of the bond.

solution. $N = 7.5$, $m = 2$, $n = 15$, $\alpha = 14\%$, $F = 2,500$

$I = 7.2\%$, $j = .036$, discount = 238.12

each end of half year coupon = $2500 \cdot \frac{\alpha}{2} = 175$

$$\text{discount} = 283.12 = C - P$$

$$C - 283.12 = P$$

but also

$$P = (Fr) a_{\overline{n}|j} + C v_j^n$$

$$= 175 a_{\overline{15}|.036} + C(1.036)^{-15}$$

$$\text{solving for } C = 5548.807228$$

$$\text{so } P = 5265.69.$$

6.4 other pricing formulas

Again we start with

$$\begin{aligned} P &= (Fr) a_{\overline{n}|j} + C v_j^n \\ &= (Fr) \left(\frac{1 - v_j^n}{j} \right) + C v_j^n \end{aligned}$$

If we use $Fr = Gj$ we see that

$$P = G - G v_j^n + C v_j^n = (C - G) v_j^n + G$$

"Basic Amount Formula"

If instead we substitute C_g for F_r we get

$$P = C_g \left(\frac{1 - v_j^n}{j} \right) + C v_j^n = \frac{g}{j} C (1 - v_j^n) + C v_j^n$$

recall $K = C v_j^n$ so:

$$P = \frac{g}{j} (C - K) + K$$

↳ Makeham's Formula

Problem: A \$2,000 bond with coupon rate of 10% payable quarterly is redeemable after an unspecified number of periods for \$2,250. The bond is bought to yield 8% convertible quarterly. If the present value of the redemption is \$869.71, find the purchase price.

Solution. $F = 2000$, $\alpha = 10\%$, $m = 4$
 $C = 2250$, $I = 8\%$, $j = 2\%$
 $K = 869.71$ Find P .

$$r = \frac{10\%}{4} = 2.5\% \quad \text{so } Fr = Cg \Rightarrow g = .022$$

so by makeham

$$P = \frac{.02}{.02} (2250 - 869.71) + 869.71 = 2403.37$$

Note if we didn't have makeham we'd have to find n first

$$\frac{C}{K} = (1+j)^n \quad \text{so} \quad n = \frac{\ln(\frac{C}{K})}{\ln(1+j)} = 48$$

calculators
killed
Makeham

Problem: An eight-year bond has annual coupons and a redemption value of \$2,338. It is purchased to yield 9% and each coupon is for \$63. Use the base amount formula to calculate the price of the bond.

solution. $N=8$, $C=2,338$, $I=9\%$, coupon = 63
Find P . $m=1$ $j=9\%$.

$$P = (Fr) a_{\overline{n}|j} + Cv_j^n$$

we'd need F and r .

by Base amount:

$$P = (C - G)v_j^n + G \quad \text{we need } G$$

$$63 = Gj \quad \text{so} \quad G = 700$$

$$\begin{aligned} P &= (2338 - 700)(1.09)^{-8} + 700 \\ &= 1522.056966. \end{aligned}$$

6.5 Bond Ammortization Schedules

Remember from Chapter 5:

L : loan

i : effective interest rate

Pmt : payment

n : # of interest periods

$OLB(t)$ = loan balance
after pmt at
time t

$I(t)$ = interest in t^{th}
pmt

$Pr(t)$ = principal in t^{th}
pmt.

$$I(t) = i OLB(t-1)$$

$$Pr(t) = Pmt - I(t)$$

$$Pr(t) = OLB(t) - OLB(t-1)$$

Now we think of a bond with price p as a loan with payments being the coupons and redemption amount.

$$L \rightarrow P$$

$$i \rightarrow j$$

$$Pmt \rightarrow Fr \quad (\text{or } Fr + C \text{ in final pmt})$$

$$n \rightarrow n$$

$$OLB(t) \rightarrow B(t) \quad \underline{\underline{\text{Book Value}}}$$

and then we still have $I(t)$ and $Pr(t)$.

using the prospective method:

$$B(t) = Fr a_{\overline{n-t}|j} + C v_j^{n-t} \quad \leftarrow \begin{array}{l} \text{standard} \\ \text{pricing} \\ \text{formula} \end{array}$$

$$B(t) = C + C a_{\overline{n-t}|j} (g - j) \quad \leftarrow \begin{array}{l} \text{premium-} \\ \text{discount} \\ \text{pricing} \\ \text{formula} \end{array}$$

Now we can derive formulas for $I(t)$ and $Pr(t)$

$$I(t) = j B(t-1) = j \left[c + c a_{\overline{n-t+1}|j} (g-j) \right]$$

$$= c j + c (1 - v_j^{n-t+1}) (g-j)$$

$$= c j + c g - c j - c g v_j^{n-t+1} + c j v_j^{n-t+1}$$

$$= c g - c v_j^{n-t+1} (g-j)$$

$$Pr(t) = B_t - B_{t-1}$$

$$= \left[c + c a_{\overline{n-t}|j} (g-j) \right] - \left[c + c a_{\overline{n-t+1}|j} (g-j) \right]$$

$$= c(g-j) \left[a_{\overline{n-t}|j} - a_{\overline{n-t+1}|j} \right]$$

$$= c(g-j) \left(\frac{1 - v_j^{n-t} - (1 - v_j^{n-t+1})}{j} \right)$$

$$= c(g-j) \left(\frac{v_j^{n-t+1} (1 - v_j^{-1})}{j} \right)$$

$$\frac{1 - \frac{1}{v}}{j}$$

$$= \frac{v-1}{v} \cdot \frac{1}{j}$$

$$= j/j = 1$$

$$= C(g-j) v_j^{n-t+1}$$

To recap: (All in terms of premium-discount)

$$B(t) = C + C a_{\overline{n-t}|j} (g-j)$$

$$I_n(t) = Cg - C v_j^{n-t+1} (g-j)$$

$$Pr(t) = C(g-j) v_j^{n-t+1}$$

To gut check we see that $I(t) + Pr(t) = Cg$ ✓

Since $C \geq 0$ and $j \geq 0 \Rightarrow v_j^{n-t+1} \geq 0$
we get that

$Pr(t) > 0$ when $g > j$ and

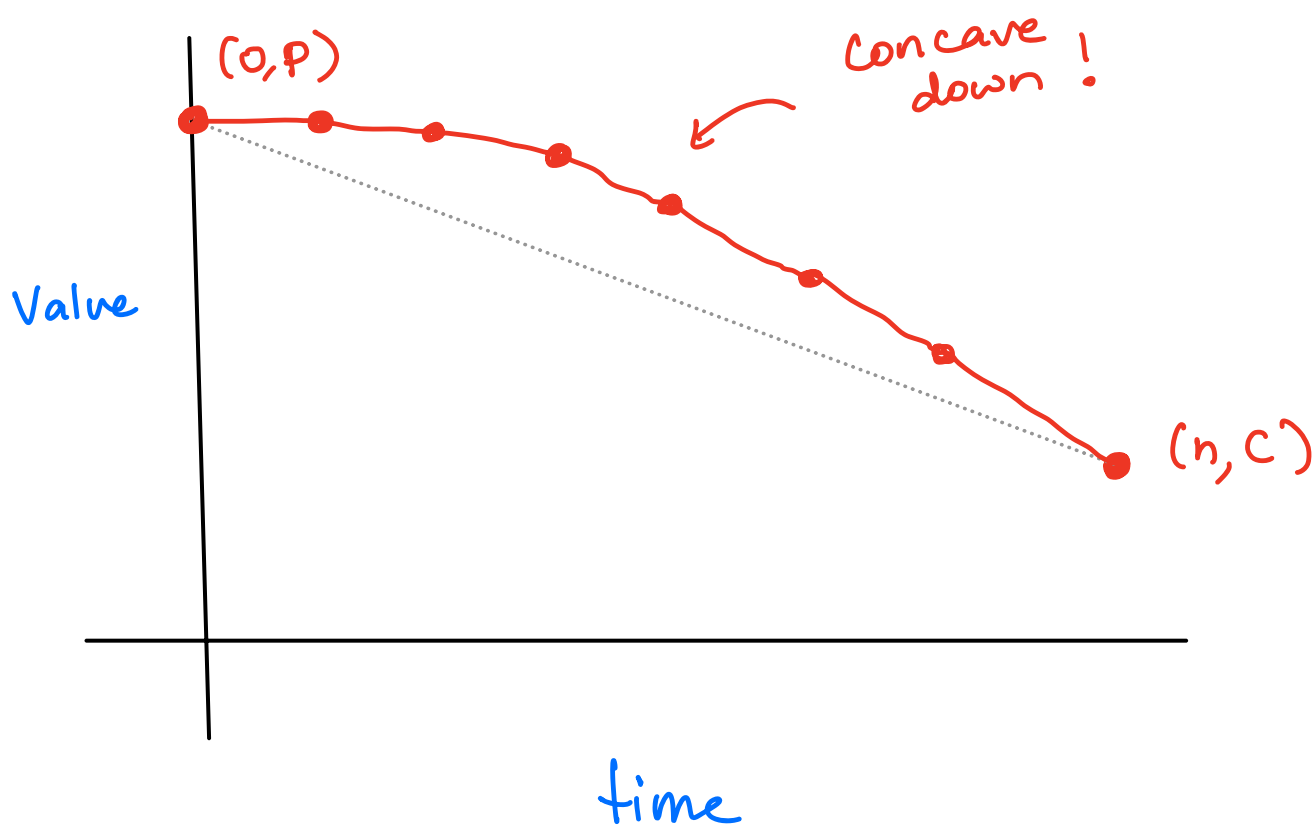
$Pr(t) < 0$ when $j > g$

(1) If a bond is sold at a premium ($g > j$) then the amount $P(t) > 0$. In this case, a part of each coupon should compensate the investor and the remainder is viewed as interest.

(2) If a bond is sold at a discount ($j > g$) then the amount $p(t) < 0$. In this case, the coupon is not sufficient to pay off all the interest due and the balance must be borrowed. Then the outstanding loan balance increases.

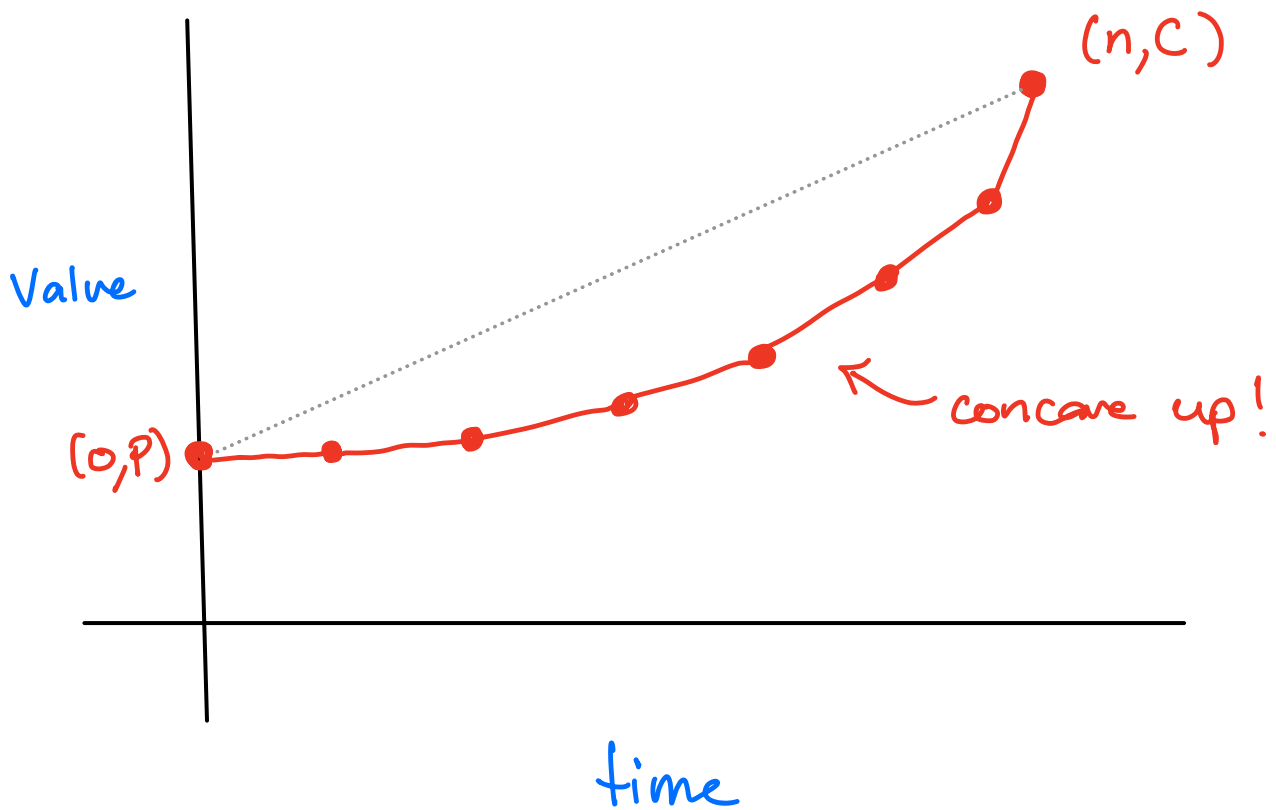
(3) If the price of the bond equals the redemption ($j = g$) then $P(t) = 0$

If case (1), Premium case $\{B(t)\}$ forms a decreasing sequence and this process is called Writing down the bond. Since $\{P(t)\}$ forms an increasing sequence we get the following picture:



We call $\{P(t)\}$ the amount for amortization of premium

In case (2), discount case $\{B(t)\}$ forms an increasing sequence and $\{-P(t)\}$ forms an increasing sequence. This process is called writing up the bond. and we get



we call $\{-P(t)\}$ the amount for accumulation of discount.

One more useful formula (Recursive)

$$B(t) = (1+j)B(t-1) - Cg$$

proof: retrospective method.

Problem: A six-year \$1,800 8.5% bond with semiannual coupons is bought for \$1,918 and is redeemable for \$1,860. Find the amount for amortization of premium at the end of two-and-a-half years and the interest due at that time.

Solution. $N=6$, $F=1800$, $\alpha=.085$, $m=2$
 $P=1918$, $C=1860$ PREMIUM

Find $Pr(2.5 \text{ years})$ and $I(2.5 \text{ years})$

$$n=12, r=.0425 \quad g = \frac{1800(.0425)}{1860} = .041129$$

$$1918 = 1800(.0425) a_{\overline{12}|j} + 1860 v_j^{12}$$

$$\text{calculator} \Rightarrow j = .0378477965$$

$$Pr(t) = C(g-j) v_j^{n-t+1}$$

$$Pr(5) = 1860(.041129 - j)(1+j)^{-(12-5+1)} = 4.53$$

$$I(5) = 76.5 - 4.53 = 71.97$$

A second solution:

remember $I(5) = B(4)j$ and

$$B(4) = Fr a_{\overline{n-4}|j} + Cv_j^{n-4} =$$

$$\Rightarrow I(5) =$$

Problem: A twelve-year bond with semiannual level coupons is bought at a premium to yield 7.5% convertible semiannually. If the amount for amortization of premium in the fourth to the last payment is \$8.02, find the premium and the total amount for amortization of premium in the first three years.

Solution. $N = 12, m = 2, I = 7.5\% \quad Pr(21) = 8.02$

Find $P - C$ and $\sum_{i=1}^6 Pr(i)$.

$$8.02 = C \left(g - \frac{.075}{2} \right) (1.0375)^{24 - 21 + 1}$$

$$\text{Premium} = P - C = \underbrace{C \left(g - \frac{.075}{2} \right)}_{\substack{11 \\ 8.02 \\ (1.0375)^4}} a_{\overline{24}|.0375} = 145.38$$

$$\begin{aligned} \sum_{i=1}^6 Pr(i) &= \sum_{i=1}^6 C \left(g - .0375 \right) V_j^{25-i} = \frac{8.02}{(1.0375)^4} \left[V_j^{24} + V_j^{23} + \dots + V_j^{19} \right] \\ &= 25.32 \end{aligned}$$

$$\text{or } \sum_{i=1}^6 \Pr(i) = B(0) - B(6) =$$

6.9 Callable Bonds

A callable bond allows the issuer to redeem the bond (or portion) at any of some fixed dates called call dates

At each call date $\rightarrow$ a redemption amt is specified.

From the buyers perspective a callable bond is less desirable because of the flexibility for the issuer. Thus, there must be some way to compensate for this for the buyer:

(1.) Higher yield rates

(2.) Redemption at callable dates is often greater than redemption at maturity

This difference is called a call premium

1 single Call Date : European option
 Multiple Call Dates : Bermuda option
 (often at coupon dates)

Any call Date : American Option

Problem: Callable Corporation is offering a \$1,000 bond with semiannual coupons and a nominal coupon rate of 8% convertible semiannually. It is a five-year bond with call dates at the end of years 2 and 3. If the redemption value at each of these dates and also at maturity is \$1,060 and the price is \$1,022, find the possible yields to the investor and note which is least.

Solution. $F = 1000$, $m = 2$, $\alpha = 8\%$, $N = 5$
 $n = 10$, $r = 4\%$, $\text{coupon} = 40$, $P = 1022$
 $C = 1060$

call date: EOY 2

$$1022 = 40 a_{\overline{4}|j} + 1060 (1+j)^{-4} \quad \leftarrow \text{biggest yield}$$

$$j = 4.77939503$$

call date: EOY 3

$$1022 = 40 a_{\overline{6}|j} + 1060 (1+j)^{-6}$$

$$j = 4.4679\%$$

maturity date: EOY 5

$$1022 = 40 a_{\overline{10}|j} + 1060 (1+j)^{-10} \quad \leftarrow \text{smallest yield}$$

$$j = 4.22043\%$$

Notice in the previous example, the bond was bought at a discount ($P < C$) and the longer the wait to redeem, the lower the yield.

This should make some intuitive sense since if the Bond issuers call the redemption short the purchaser is compensated by yield rates.

In fact:

Claim. Suppose is bought at a discount and let y_k be the yield if the bond is called at time k .

Then $y_2 > y_3 > y_4 > \dots > y_n$

proof. Fix k in $\{2, 3, 4, 5, \dots, n\}$

$$P = (cg) a_{\overline{k}|y_k} + C(1+y_k)^{-k}$$

$$= (cg) \left(a_{\overline{k-1}|y_k} + (1+y_k)^{-k} \right) + C(1+y_k)^{-k}$$

$$= (cg) a_{\overline{k-1}|y_k} + C(g+1)(1+y_k)^{-k}$$

$$= Cg a_{\overline{k-1}|y_k} + C \left(\frac{1+g}{1+y_k} \right) (1+y_k)^{-(k-1)}$$

Since discount $g < y_k \Rightarrow \frac{1+g}{1+y_k} < 1$

$$\text{so } P < (cg) a_{\overline{k-1}|y_k} + C(1+y_k)^{-(k-1)}$$

but

$$P = (cg) a_{\overline{k-1}|y_{k-1}} + C(1+y_{k-1})^{-(k-1)}$$

$$\text{So } y_{k-1} > y_k$$

Problem: A \$2,000 six-year 7% bond with annual coupons is purchased for \$2,200. If held to maturity, it is redeemable at par. The issuer may redeem the bond at the end of any year once it has been held for at least three years. The call premiums are \$150 at $t = 3$ years, \$140 at $t = 4$ years, and \$130 at $t = 5$ years. Determine the possible yields to the investor and note which is minimal and which is maximal.

Solution. $F = 2000, N = 6, \alpha = 7\%, m = 1$

$$n = 6, r = 7\%, P = 2200$$

If called at $k = 6 \rightarrow C = 2000$

If called at $k = 3 \rightarrow C = 2150$

If called at $k = 4 \rightarrow C = 2140$

If called at $k = 5 \rightarrow C = 2130$

$$2200 = 140 a_{\overline{6}|j} + 2000(1+j)^{-6} \Rightarrow j = 5.028057\%$$

$$2200 = 140 a_{\overline{5}|y_5} + 2130(1+y_5)^{-5} \Rightarrow y_5 = 5.796899\%$$

$$2200 = 140 a_{\overline{4}|y_4} + 2140(1+y_4)^{-4} \Rightarrow y_4 = 5.737774\%$$

$$2200 = 140 a_{\overline{3}|y_3} + 2150(1+y_3)^{-3} \Rightarrow y_3 = 5.64727\%$$

Problem: A \$1,000 10% ten-year bond with annual coupons is purchased for \$1,100. Its redemption amount at maturity is \$980 while it may be called in at the end of any year, the call premiums being \$115 at the ends of years 1-3, \$80 at the ends of years 4-6, and \$40 at the ends of years 7-9. Find the minimum yield rate to the purchaser.

Solution. $F=1000, \alpha=10\%, N=10, m=1$

$n=10, r=10\%, \text{Coupon}=100, P=1100$

$C=980$

call at EOY 1-3 $= 115 + 980 = 1095$

call at EOY 4-6 $= 80 + 980 = 1060$

call at EOY 7-9 $= 40 + 980 = 1020$

Min yield rate

EOY 1: 8.7272%

EOY 4: 8.287342%

EOY 7: 8.2828666% ← smallest

EOY 10: 8.350%